

Benchmark Results

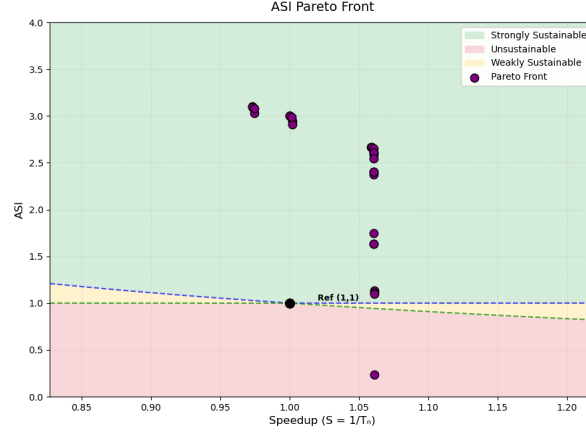
Contents

1	ML2 benchmark	2
1.1	Greedy	2
1.2	SPEA2 - COLE	3
1.2.1	hyperparameter settings 1	3
1.3	Max-value Entropy Search for Multi-objective Bayesian Optimization (MESMO) . .	3
1.3.1	hyperparameter settings 1	3
2	ML2 and CCl benchmarks	6
2.1	Greedy	6
2.2	SPEA2 - COLE	6
2.2.1	hyperparameter settings 1	6
2.3	MESMO	7
2.3.1	hyperparameter settings 1	7
2.3.2	hyperparameter settings 2	8
3	ML2, CCl, MIP and EI benchmarks	11
3.1	MESMO	11
3.1.1	hyperparameter settings 1	11
3.1.2	hyperparameter settings 2	12
3.1.3	hyperparameter settings 3	12
3.2	SPEA2 - COLE	14
3.2.1	Hyperparameter settings 1	14
4	ML2 benchmark, larger working set	15
4.1	Reference: L3 size = 8 MiB, L2 size = 0.25 MiB	15
4.2	Reference: L3 size = 2 MiB, L2 size = 0.25 MiB	16
5	Benchmarks	16
5.1	ML2	16
5.2	CCl	17
5.3	MIP	18
5.4	EI	19

1 ML2 benchmark

ML2: L2 resident (depending on LFSR settings) linked list traversal

1.1 Greedy



Result:

Total sniper runs: 781

Final hypervolume: 3.2594

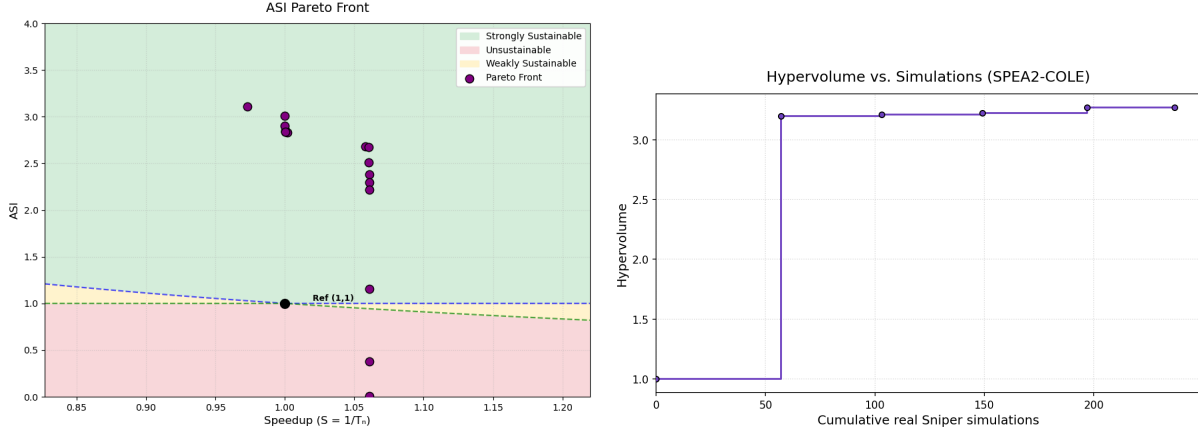
Front:

bpt	bp	llda	lld	llia	lli	l2a	l2	l3a	l3	nhist	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53		4	64				512		2048	4			1024	0.2323	1.0610	67.94	141.87	Unsustainable
pentium_m		4	64	8	16		512		1024		8	2	1024	1.1008	1.0610	49.33	34.59	Strongly Sust.
pentium_m		4	64	8			512	4	1024		2	2	1024	1.1154	1.0609	47.61	34.56	Strongly Sust.
pentium_m		4	64				512		1024			2	1024	1.1342	1.0608	47.96	33.90	Strongly Sust.
a53			64				512		2048		2			1.6355	1.0608	55.79	22.20	Strongly Sust.
a53	512		64				512		2048		2			1.6355	1.0608	55.79	22.20	Strongly Sust.
a53			64				512		2048				16	1.7450	1.0607	55.19	20.90	Strongly Sust.
a53	512		64				512		1024			2		2.3749	1.0607	47.80	16.21	Strongly Sust.
pentium_m		4	64	8			512		1024		2	64		2.4038	1.0607	45.04	16.33	Strongly Sust.
a53	512	4	64	8			512		1024		2	64		2.4038	1.0606	45.04	16.33	Strongly Sust.
a53	512	4	64				512		1024		2	64		2.5447	1.0606	43.32	15.61	Strongly Sust.
a53		4	64				512	8	1024		2	64		2.5897	1.0606	41.24	15.56	Strongly Sust.
a53	512	4	64				512		1024		2	16		2.6083	1.0606	43.17	15.25	Strongly Sust.
a53		4	64				512		1024		2	16		2.6083	1.0605	43.17	15.25	Strongly Sust.
a53		4	64				512	8	1024		2	16		2.6546	1.0605	41.09	15.19	Strongly Sust.
a53		4					512	8	1024		2	16		2.6666	1.0589	40.92	15.14	Strongly Sust.
a53		4					512	4	1024		2	16		2.6682	1.0589	40.92	15.13	Strongly Sust.
a53		4	64			4			1024		2	32		2.9076	1.0019	36.49	14.31	Strongly Sust.
a53		4	64			4			1024		2	16		2.9477	1.0018	36.40	14.12	Strongly Sust.
a53		4	64					8	1024		2	16		2.9861	1.0013	34.86	14.08	Strongly Sust.
a53		4						8	1024		2	16		3.0002	0.9999	34.69	14.03	Strongly Sust.
a53		4						4	1024		2	16		3.0022	0.9999	34.69	14.02	Strongly Sust.
a53		4	64			4	128		1024		2	16		3.0325	0.9746	35.43	13.81	Strongly Sust.
a53		4	64				128	8	1024		2	16		3.0823	0.9744	33.37	13.77	Strongly Sust.
a53		4					128	8	1024		2	16		3.0970	0.9732	33.20	13.72	Strongly Sust.
a53		4					128	4	1024		2	16		3.0991	0.9731	33.20	13.71	Strongly Sust.

1.2 SPEA2 - COLE

1.2.1 hyperparameter settings 1

Hyperparameter settings: Patience = 1 (amount of iterations to wait before stopping after no improvement), max_iterations: int = 10. (max iter never reached)



Result:

Total sniper runs: 237

Final hypervolume: 3.2693

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	1024	4	64	8	16	8	512	16	1024	3	2	8	1024	0.0070	1.0608	157.56	1188.00	Unsustainable
pentium_m		8	64	4	16	8	512	16	1024		2	4	512	0.3748	1.0608	57.72	95.43	Unsustainable
pentium_m		4	64	4	16	8	512	4	1024		2	2	1024	1.1558	1.0608	45.64	33.82	Strongly Sust.
pentium_m		4	64	8	64	4	512	8	2048		4	2	256	2.2207	1.0607	48.56	17.24	Strongly Sust.
tage		4	64	8	16	8	512	4	2048		10	2	16	2.2940	1.0607	49.42	16.59	Strongly Sust.
pentium_m		4	64	4	32	8	512	16	2048		4	2	32	2.3800	1.0606	48.12	16.14	Strongly Sust.
pentium_m		4	64	8	64	8	512	4	1024		10	2	16	2.5075	1.0605	43.13	15.86	Strongly Sust.
tage		4	64	4	16	4	512	4	1024		8	2	16	2.6748	1.0605	40.76	15.11	Strongly Sust.
tage		4	16	4	32	8	512	4	1024		4	2	16	2.6778	1.0581	40.73	15.10	Strongly Sust.
pentium_m		4	64	8	64	4	256	8	1024		2	2	16	2.8286	1.0018	36.35	14.72	Strongly Sust.
pentium_m		4	32	8	64	4	256	8	1024		10	2	16	2.8417	1.0004	36.18	14.67	Strongly Sust.
pentium_m		4	32	8	64	4	256	8	1024		10	2	16	2.8417	1.0004	36.18	14.67	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024		10	2	16	2.8550	1.0003	35.52	14.66	Strongly Sust.
pentium_m		4	32	4	16	8	256	4	1024		10	2	128	2.9034	1.0000	34.66	14.50	Strongly Sust.
pentium_m		4	32	4	16	8	256	4	1024		2	2	16	3.0118	0.9999	34.45	14.00	Strongly Sust.
pentium_m		4	32	4	16	8	128	4	1024		2	2	16	3.1091	0.9731	32.96	13.69	Strongly Sust.

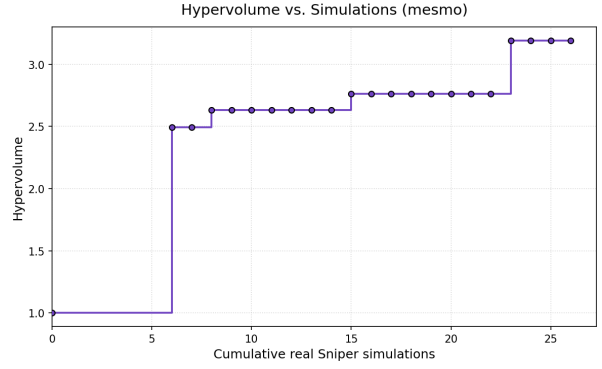
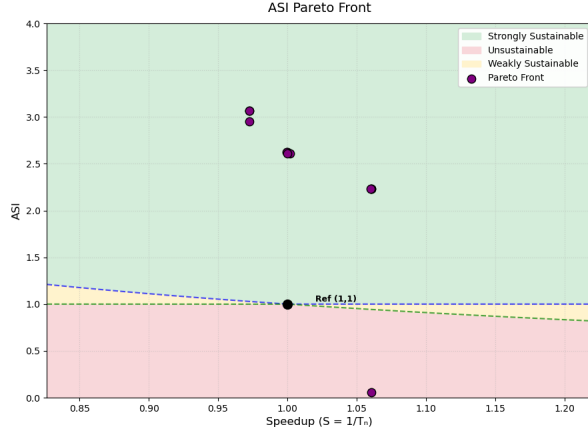
1.3 Max-value Entropy Search for Multi-objective Bayesian Optimization (MESMO)

source: <https://arxiv.org/pdf/2110.06980>

1.3.1 hyperparameter settings 1

Hyperparameter settings: 400 candidate samples (400 candidate configurations are considered per iteration; each is scored by the information gained, averaged over a default of 10 Monte Carlo posterior-function samples (amount is hyperparameter, default is 10), and the top-scoring one (also hyperparameter, default is 1) is evaluated.) are used and 7 initial configurations (including baseline, always the case!) have been run.

- 20 iterations



Result:

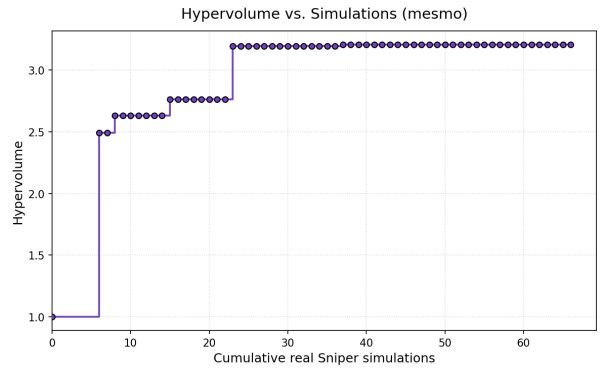
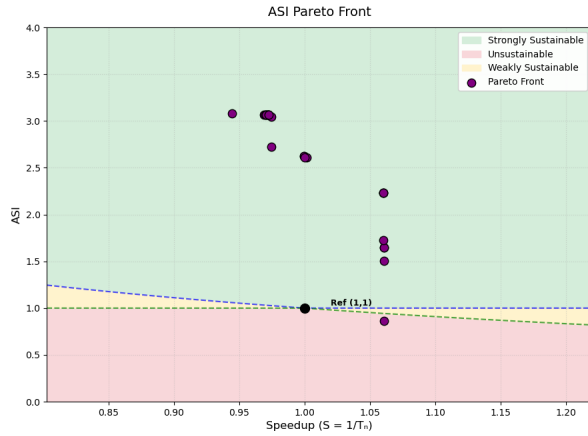
Total sniper runs: 26

Final hypervolume: 3.1908

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	8	16	8	512	16	1024				10	8	256	0.0537	1.0608	83.61	533.35	Unsustainable
a53	512	8	64	8	16	8	512	16	1024	2			10	2	256	2.2291	1.0606	49.29	17.09	Strongly Sust.
pentium_m		8	64	8	16	8	512	16	1024				10	2	256	2.2292	1.0605	49.29	17.08	Strongly Sust.
nn		8	64	8	16	8	512	16	1024		16	0.005	10	2	256	2.2293	1.0602	49.29	17.08	Strongly Sust.
a53	512	4	64	4	16	4	256	4	2048	4			2	2	256	2.6064	1.0018	41.36	15.45	Strongly Sust.
nn		4	64	4	16	4	256	4	2048		16	0.0005	2	2	256	2.6071	1.0001	41.36	15.44	Strongly Sust.
pentium_m		4	16	4	16	4	256	4	2048				2	2	256	2.6273	0.9999	40.99	15.36	Strongly Sust.
a53	512	4	16	4	16	4	256	4	2048	4			2	2	256	2.6273	0.9997	40.99	15.36	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			2	2	256	2.9545	0.9724	33.33	14.37	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			4	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			2	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.

- 60 iterations



Result:

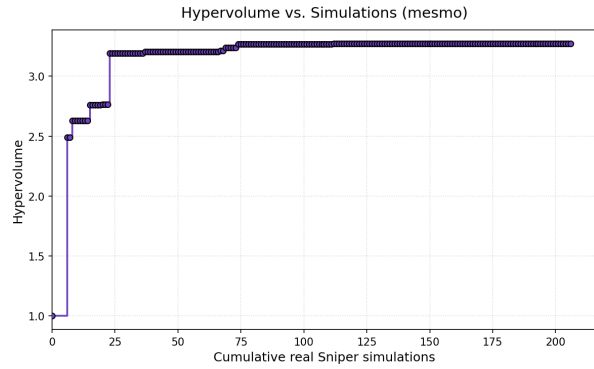
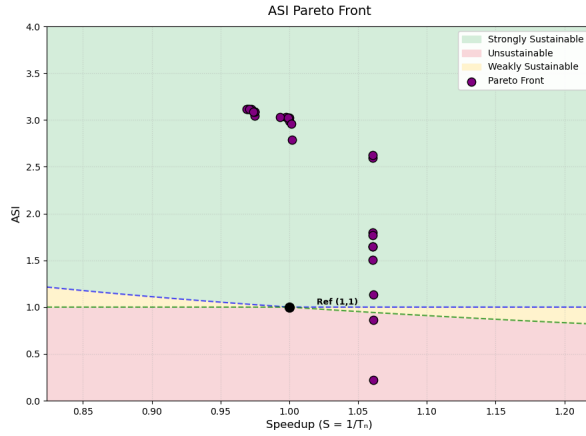
Total sniper runs: 66

Final hypervolume: 3.2056

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robv	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m	8	64	8	16	4	512	8	4096					10	2	1024	0.8626	1.0609	70.94	37.26	Unsustainable
pentium_m	8	64	8	16	8	512	16	1024					10	2	512	1.5034	1.0608	50.78	25.06	Strongly Sust.
pentium_m	4	64	8	64	8	512	16	2048					10	4	128	1.6488	1.0608	53.40	22.41	Strongly Sust.
pentium_m	4	64	8	64	8	512	16	2048					8	4	128	1.6488	1.0607	53.40	22.41	Strongly Sust.
a53	512	8	64	8	16	8	512	16	4096	2			10	2	256	1.7280	1.0606	66.19	19.35	Strongly Sust.
a53	512	8	64	8	16	8	512	16	1024	2			10	2	256	2.2291	1.0606	49.29	17.09	Strongly Sust.
pentium_m	8	64	8	16	8	512	16	1024					10	2	256	2.2292	1.0605	49.29	17.08	Strongly Sust.
nn	8	64	8	16	8	512	16	1024		16	0.005		10	2	256	2.2293	1.0602	49.29	17.08	Strongly Sust.
a53	512	4	64	4	16	4	256	4	2048	4			2	2	256	2.6064	1.0018	41.36	15.45	Strongly Sust.
nn	4	64	4	16	4	256	4	2048		16	0.0005		2	2	256	2.6071	1.0001	41.36	15.44	Strongly Sust.
pentium_m	4	16	4	16	4	256	4	2048					2	2	256	2.6273	0.9999	40.99	15.36	Strongly Sust.
a53	512	4	16	4	16	4	256	4	2048	4			2	2	256	2.6273	0.9997	40.99	15.36	Strongly Sust.
tage	4	64	4	32	4	128	16	2048					4	2	64	2.7216	0.9747	40.44	14.89	Strongly Sust.
nn	4	64	4	32	8	128	4	1024		64	0.0005		8	2	32	3.0414	0.9746	33.46	13.95	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	3			8	2	32	3.0671	0.9725	33.09	13.87	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			10	2	32	3.0671	0.9724	33.09	13.87	Strongly Sust.
tage	4	16	4	32	8	128	4	1024					2	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.
a53	512	4	16	4	32	8	128	4	1024	4			8	2	32	3.0672	0.9723	33.09	13.87	Strongly Sust.
nn	4	16	4	32	8	128	4	1024		16	0.0005		10	2	32	3.0675	0.9717	33.09	13.87	Strongly Sust.
nn	4	16	4	32	8	128	4	1024		64	0.005		2	2	32	3.0681	0.9706	33.09	13.86	Strongly Sust.
nn	4	16	4	32	8	128	4	1024		64	0.005		10	2	32	3.0682	0.9703	33.09	13.86	Strongly Sust.
nn	4	16	4	32	8	128	4	1024		32	0.0005		10	2	32	3.0684	0.9700	33.09	13.86	Strongly Sust.
nn	4	16	4	32	8	128	4	1024		32	0.0005		8	2	32	3.0688	0.9692	33.09	13.86	Strongly Sust.
nn	4	16	4	32	8	128	4	1024		64	0.0005		8	2	32	3.0814	0.9447	33.09	13.80	Strongly Sust.

- 200 iterations



Result:

Total sniper runs: 206

Final hypervolume: 3.2729

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	64	8	512	4	2048				8	4	1024	0.2219	1.0609	72.96	142.35	Unsustainable
pentium_m		8	64	8	16	4	512	8	4096				10	2	1024	0.8626	1.0609	70.94	37.26	Unsustainable
pentium_m		4	64	4	16	8	512	16	1024				2	2	1024	1.1369	1.0609	47.72	33.88	Strongly Sust.
pentium_m		8	64	8	16	8	512	16	1024				10	2	512	1.5034	1.0608	50.78	25.06	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	2048				10	4	128	1.6488	1.0608	53.40	22.41	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	2048				8	4	128	1.6488	1.0607	53.40	22.41	Strongly Sust.
pentium_m		8	64	8	64	8	512	16	4096				8	2	32	1.7692	1.0606	66.61	18.84	Strongly Sust.
pentium_m		4	64	8	32	8	512	16	1024				8	4	64	1.7965	1.0606	48.05	21.39	Strongly Sust.
pentium_m		4	64	4	16	8	512	8	1024				10	2	64	2.5978	1.0606	41.00	15.54	Strongly Sust.
nn		4	64	4	32	8	512	4	1024		64	0.0005	8	2	32	2.6225	1.0606	41.18	15.37	Strongly Sust.
tage		8	64	4	32	4	256	8	1024				2	2	32	2.7856	1.0019	38.83	14.70	Strongly Sust.
a53	2048	4	64	4	64	8	256	8	1024	2			4	2	16	2.9603	1.0013	35.45	14.15	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		16	0.005	2	2	16	2.9891	1.0002	34.74	14.08	Strongly Sust.
nn		4	32	4	16	4	256	4	1024		16	0.005	10	2	16	3.0249	1.0002	33.91	13.99	Strongly Sust.
nn		4	16	4	32	4	256	4	1024		16	0.005	10	2	16	3.0264	0.9993	33.96	13.97	Strongly Sust.
nn		4	16	4	32	4	256	4	1024		64	0.005	10	2	16	3.0272	0.9977	33.96	13.97	Strongly Sust.
nn		4	32	4	16	4	256	4	1024		16	0.0005	10	2	16	3.0284	0.9932	33.91	13.97	Strongly Sust.
a53	2048	4	64	4	32	4	128	4	1024	4			8	2	32	3.0444	0.9746	33.43	13.94	Strongly Sust.
a53	2048	4	64	4	32	4	128	4	1024	4			8	2	16	3.0873	0.9746	33.35	13.75	Strongly Sust.
a53	512	4	64	4	32	4	128	4	1024	4			8	2	16	3.0873	0.9746	33.35	13.75	Strongly Sust.
nn		4	64	4	32	4	128	4	1024		16	0.0005	8	2	16	3.0877	0.9738	33.35	13.75	Strongly Sust.
tage		4	16	4	32	4	128	4	1024				10	2	16	3.1136	0.9724	32.98	13.67	Strongly Sust.
nn		4	16	4	32	4	128	4	1024		16	0.001	10	2	16	3.1144	0.9709	32.98	13.67	Strongly Sust.
nn		4	16	4	32	4	128	4	1024		64	0.005	10	2	16	3.1149	0.9699	32.98	13.66	Strongly Sust.
nn		4	16	4	32	4	128	4	1024		64	0.001	10	2	16	3.1153	0.9692	32.98	13.66	Strongly Sust.

HV was already 3.2727 at iter 152, before that also quite slow increase. Amount of points did increase with 10.

2 ML2 and CCI benchmarks

ML2: L2 resident (depending on LFSR settings) linked list traversal

CCI: Impossible control with large basic blocks (potentially larger penalty)

2.1 Greedy

Became infeasible with new parameterspace.

first iteration: 33 simulations

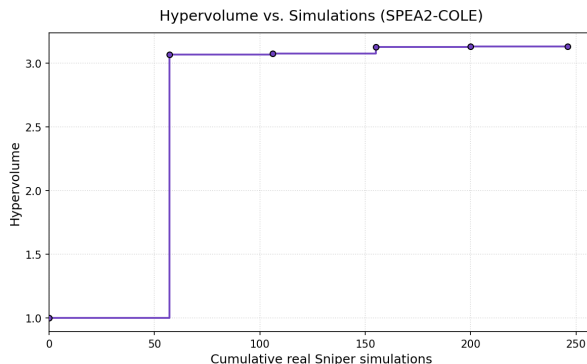
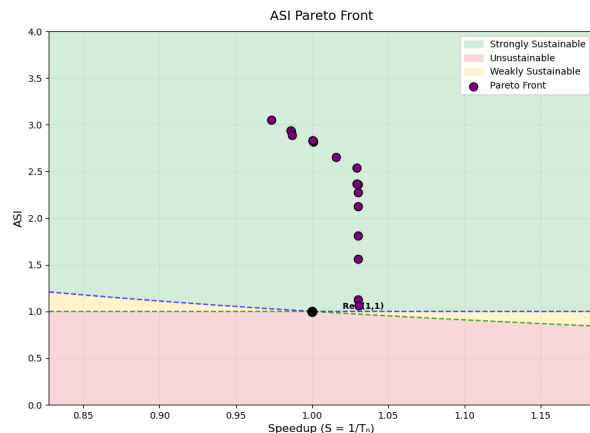
second iteration: ≥ 100 simulations

2.2 SPEA2 - COLE

2.2.1 hyperparameter settings 1

Hyperparameter settings: Patience = 1 (amount of iterations to wait before stopping after no improvement), max_iterations: int = 30,

following parameters are default: num_populations: int = 3, population_size: int = 20, archive_size: int = 10, p_mutation: float = 0.10, p_crossover: float = 0.90, p_migration: float = 0.10.



Result:

Total sniper runs: 246

Final hypervolume: 3.1294

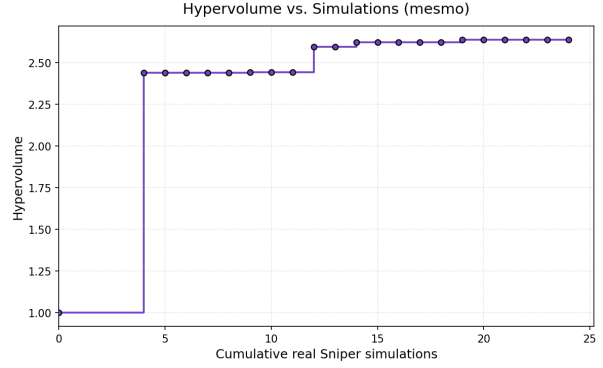
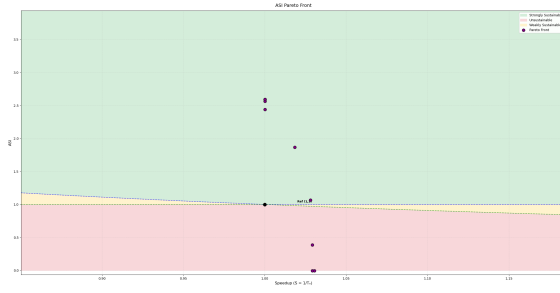
Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robv	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	1024	4	64	8	64	8	512	8	2048	3			4	2	1024	1.0624	1.0304	53.11	36.02	Strongly Sust.
a53	1024	4	64	8	64	4	512	8	1024	3			4	2	1024	1.1281	1.0303	47.82	35.26	Strongly Sust.
pentium_m		4	64	4	16	4	512	4	2048				2	2	512	1.5617	1.0302	49.53	25.16	Strongly Sust.
pentium_m		4	64	8	64	8	512	16	4096				10	2	256	1.8130	1.0302	62.43	19.65	Strongly Sust.
pentium_m		8	64	4	64	8	512	16	2048				4	2	256	2.1272	1.0302	53.37	17.96	Strongly Sust.
pentium_m		4	64	4	64	8	512	4	2048				2	2	256	2.2773	1.0302	48.96	17.32	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	1024				2	2	256	2.3633	1.0301	45.75	17.08	Strongly Sust.
pentium_m		8	32	4	32	8	512	4	1024				4	2	256	2.3675	1.0293	45.51	17.08	Strongly Sust.
pentium_m		4	32	4	16	4	512	4	1024				2	2	256	2.5419	1.0293	40.92	16.42	Strongly Sust.
pentium_m		4	32	4	32	8	512	8	1024				4	2	16	2.6494	1.0158	40.92	15.75	Strongly Sust.
a53	512	4	64	4	16	4	256	4	1024	2			4	2	256	2.8179	1.0009	34.41	15.47	Strongly Sust.
pentium_m		4	32	4	16	4	256	4	1024				4	2	256	2.8307	1.0005	34.23	15.41	Strongly Sust.
pentium_m		4	32	4	16	4	256	4	1024				4	2	256	2.8307	1.0005	34.23	15.41	Strongly Sust.
pentium_m		4	32	4	32	8	128	8	1024				4	2	256	2.8878	0.9868	33.52	15.18	Strongly Sust.
a53	512	4	32	4	32	8	128	8	1024	3			4	2	128	2.9270	0.9865	33.40	14.99	Strongly Sust.
a53	1024	4	16	4	32	8	128	8	1024	3			4	2	128	2.9384	0.9861	33.21	14.95	Strongly Sust.
pentium_m		4	16	4	32	8	128	8	1024				4	2	16	3.0553	0.9734	33.00	14.39	Strongly Sust.

2.3 MESMO

2.3.1 hyperparameter settings 1

hyperparameter settings: 200 candidate samples (200 candidate configurations are considered per iteration; each is scored by the information gained, averaged over a default of 10 Monte Carlo posterior-function samples (amount is hyperparameter, default is 10), and the top-scoring one (also hyperparameter, default is 1) is evaluated.) are used and 5 initial configurations (including baseline, always the case!) have been run. 20 iterations.



Result:

Total sniper runs: 24

Final hypervolume: 2.6384

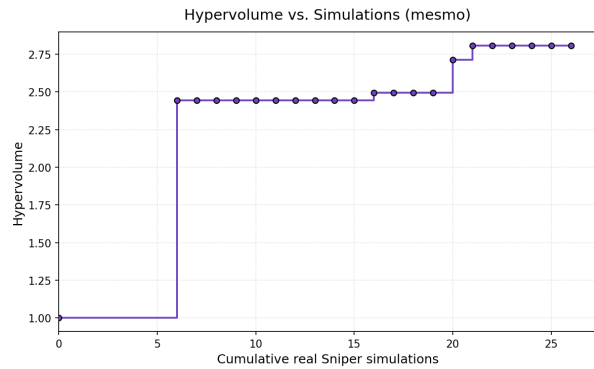
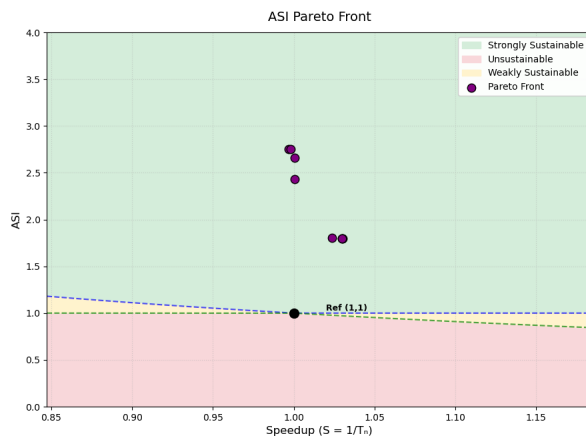
Front:

bpt	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m	8	64	4	16	8	512	4	1024			8	10	1024	-0.0056	1.0305	239.58	2631.93	Unsustainable
pentium_m	8	16	4	16	8	512	4	1024			8	10	1024	-0.0054	1.0293	238.13	2631.81	Unsustainable
tage	4	32	4	64	8	512	8	2048			8	4	512	0.3855	1.0293	57.08	96.36	Unsustainable
nn	8	16	4	16	8	512	8	2048	32	0.0005	2	2	1024	1.0671	1.0281	53.81	35.68	Strongly Sust.
tage	4	32	4	64	8	512	8	2048			8	4	16	1.8672	1.0185	49.44	21.06	Strongly Sust.
pentium_m	4	16	8	16	4	256	4	2048			2	2	256	2.4405	1.0001	42.61	16.90	Strongly Sust.
pentium_m	4	16	4	64	4	256	4	2048			2	2	256	2.5637	1.0001	41.82	16.18	Strongly Sust.
pentium_m	4	16	4	16	4	256	4	2048			8	2	256	2.5931	1.0001	40.99	16.09	Strongly Sust.
pentium_m	4	16	4	16	4	256	4	2048			2	2	256	2.5931	1.0001	40.99	16.09	Strongly Sust.

2.3.2 hyperparameter settings 2

hyperparameter settings: 400 candidate samples are used and 7 initial configurations have been run.

- 20 iterations



Result:

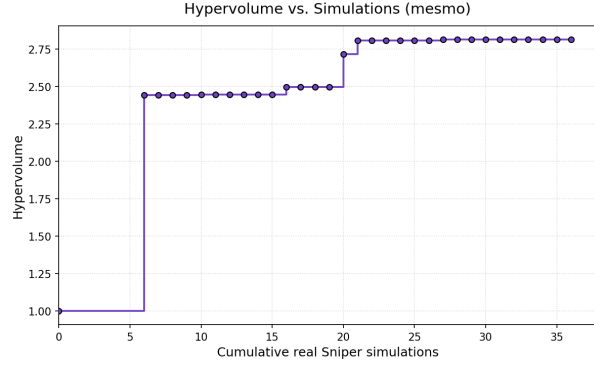
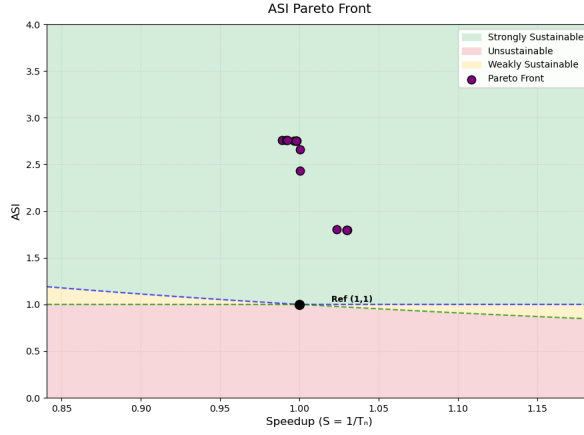
Total sniper runs: 26

Final hypervolume: 2.8076

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region	
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.	
a53	512	8	64	4	32	4	512	4	4096	4			2	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.	
a53	512	8	64	4	32	4	512	4	4096	4			10	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.	
nn		8	64	4	32	4	512	4	4096		16	0.001	2	2	256	1.8013	1.0235	65.13	19.35	Strongly Sust.	
pentium_m	4		32	8		16	4	256	4	2048			2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.	
pentium_m	4		32	8		16	4	256	4	1024			2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.	
nn	4		32	8		16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
pentium_m	4		32	8		16	4	256	4	1024			2	2	32	2.7531	0.9967	35.61	15.71	Strongly Sust.	

- 30 iterations



Result:

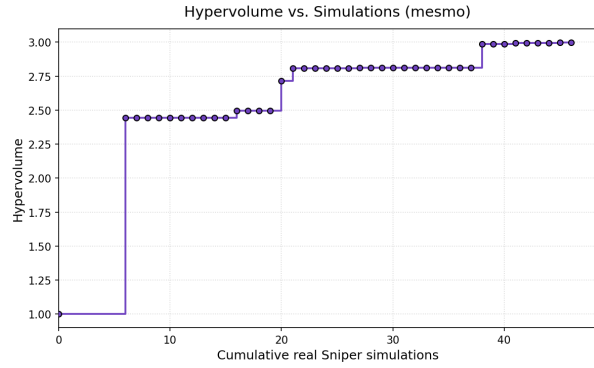
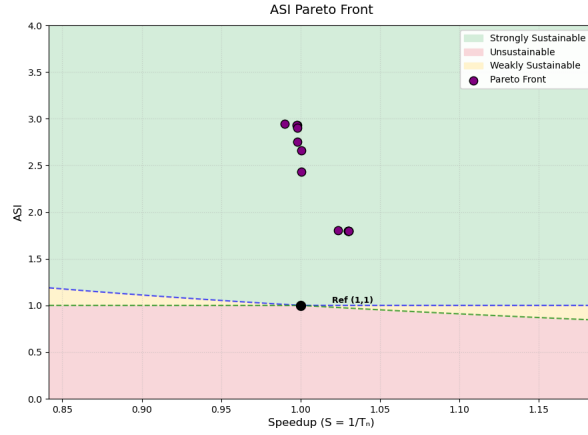
Total sniper runs: 36

Final hypervolume: 2.8132

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			2	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			10	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
nn		8	64	4	32	4	512	4	4096		16	0.001	2	2	256	1.8013	1.0235	65.13	19.35	Strongly Sust.
pentium_m	4	32	8	16	4	256	4	2048					2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m	4	32	8	16	4	256	4	1024					2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn	4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.	
a53	2048	4	32	8	16	4	256	4	1024	4			2	2	32	2.7520	0.9980	35.61	15.71	Strongly Sust.
tage	4	32	8	16	4	256	4	1024					2	2	32	2.7522	0.9978	35.61	15.71	Strongly Sust.
pentium_m	4	32	8	16	4	256	4	1024					2	2	32	2.7531	0.9967	35.61	15.71	Strongly Sust.
a53	1024	4	32	8	16	4	256	4	1024	2			2	2	32	2.7567	0.9923	35.61	15.68	Strongly Sust.
nn	4	32	8	16	4	256	4	1024			64	0.0005	2	2	32	2.7573	0.9915	35.61	15.68	Strongly Sust.
nn	4	32	8	16	4	256	4	1024			64	0.005	2	2	32	2.7587	0.9892	35.61	15.67	Strongly Sust.

- 40 iterations



Result:

Total sniper runs: 46

Final hypervolume: 2.9982

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robcb	robdb	robwb	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	32	4	512	4	4096				8	2	256	1.7997	1.0302	65.13	19.37	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			2	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
a53	512	8	64	4	32	4	512	4	4096	4			10	2	256	1.7998	1.0298	65.13	19.37	Strongly Sust.
nn		8	64	4	32	4	512	4	4096		16	0.001	2	2	256	1.8013	1.0235	65.13	19.35	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	32	2.8998	0.9981	34.82	14.99	Strongly Sust.
a53	2048	4	32	4	16	4	256	4	1024	4			2	2	32	2.9337	0.9980	34.00	14.89	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	4			2	2	32	2.9341	0.9975	34.00	14.89	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	16	2.9440	0.9902	34.74	14.77	Strongly Sust.

- 50 iterations

Mistake with automatic plotting, but conclusion is that hypervolume keeps increasing.

Result:

Total sniper runs: 56

Final hypervolume: 3.0285

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robcb	robdb	robwb	ASI	Speedup	Area	PeakPow	Region
tage		8	64	8	16	4	512	4	4096				8	2	1024	0.8705	1.0304	70.96	38.15	Unsustainable
pentium_m		8	64	8	32	8	512	16	2048				10	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				8	2	256	1.7997	1.0302	65.13	19.37	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	128	2.0450	1.0299	54.39	18.54	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	32	2.8998	0.9981	34.82	14.99	Strongly Sust.
a53	2048	4	32	4	16	4	256	4	1024	4			2	2	32	2.9337	0.9980	34.00	14.89	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	4			2	2	32	2.9341	0.9975	34.00	14.89	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	16	2.9440	0.9902	34.74	14.77	Strongly Sust.
pentium_m		4	32	4	64	4	256	4	1024				2	2	16	2.9461	0.9874	34.74	14.76	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		64	0.001	2	2	16	2.9679	0.9577	34.74	14.65	Strongly Sust.

- 60 iterations

Result:

Total sniper runs: 66

Final hypervolume: 3.0375

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
tage		8	64	8	16	4	512	4	4096				8	2	1024	0.8705	1.0304	70.96	38.15	Unsustainable
pentium_m		8	64	8	32	8	512	16	2048				10	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	512	1.4025	1.0302	56.00	26.71	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				8	2	256	1.7997	1.0302	65.13	19.37	Strongly Sust.
pentium_m		8	64	4	32	4	512	4	4096				2	2	256	1.7997	1.0301	65.13	19.37	Strongly Sust.
pentium_m		8	64	8	32	8	512	16	2048				8	2	128	2.0450	1.0299	54.39	18.54	Strongly Sust.
a53	2048	8	32	4	64	4	512	16	1024	4			8	2	64	2.3621	1.0287	47.90	16.83	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	2048				2	2	256	2.4315	1.0005	42.80	16.94	Strongly Sust.
pentium_m		4	32	8	16	4	256	4	1024				2	2	256	2.6597	1.0005	35.85	16.23	Strongly Sust.
nn		4	32	8	16	4	256	4	1024		16	0.005	2	2	32	2.7519	0.9982	35.61	15.71	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	32	2.8998	0.9981	34.82	14.99	Strongly Sust.
a53	2048	4	32	4	16	4	256	4	1024	4			2	2	32	2.9337	0.9980	34.00	14.89	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	4			2	2	32	2.9341	0.9975	34.00	14.89	Strongly Sust.
a53	2048	4	32	4	64	4	256	4	1024	4			2	2	16	2.9440	0.9902	34.74	14.77	Strongly Sust.
pentium_m		4	32	4	64	4	256	4	1024				8	2	16	2.9461	0.9874	34.74	14.76	Strongly Sust.
pentium_m		4	32	4	64	4	256	4	1024				2	2	16	2.9461	0.9874	34.74	14.76	Strongly Sust.
a53	512	4	32	4	64	4	256	4	1024	4			2	2	16	2.9463	0.9871	34.74	14.76	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		16	0.001	10	2	16	2.9521	0.9789	34.74	14.73	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		16	0.001	2	2	16	2.9678	0.9580	34.74	14.65	Strongly Sust.
nn		4	32	4	64	4	256	4	1024		64	0.001	2	2	16	2.9679	0.9577	34.74	14.65	Strongly Sust.

3 ML2, CCl, MIP and EI benchmarks

ML2: L2 resident (depending on LFSR settings) linked list traversal

CCl: Impossible control with large basic blocks (potentially larger penalty)

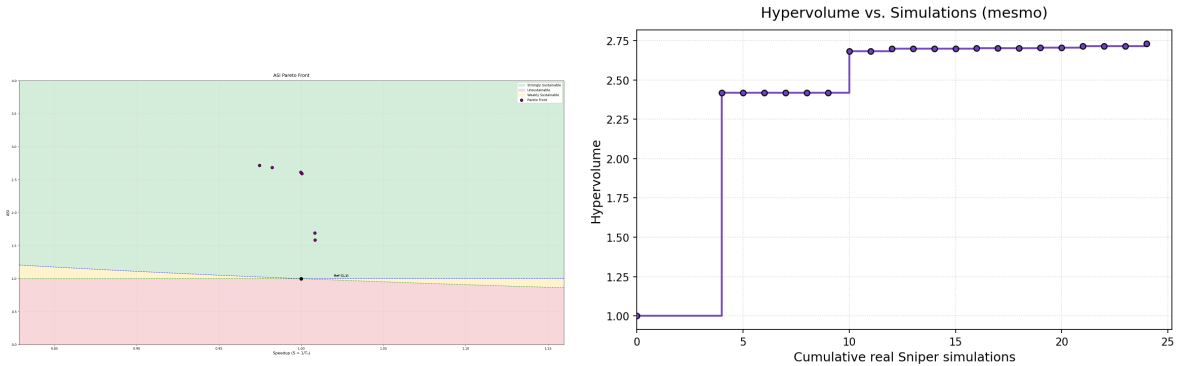
MIP: Large Instruction Region – Instruction cache Misses

EI: Integer Execution – 8 Independent computations per iteration

3.1 MESMO

3.1.1 hyperparameter settings 1

hyperparameter settings: 200 candidate samples are used and 5 initial configurations are ran. 20 iterations.

**Result:**

Total sniper runs: 24

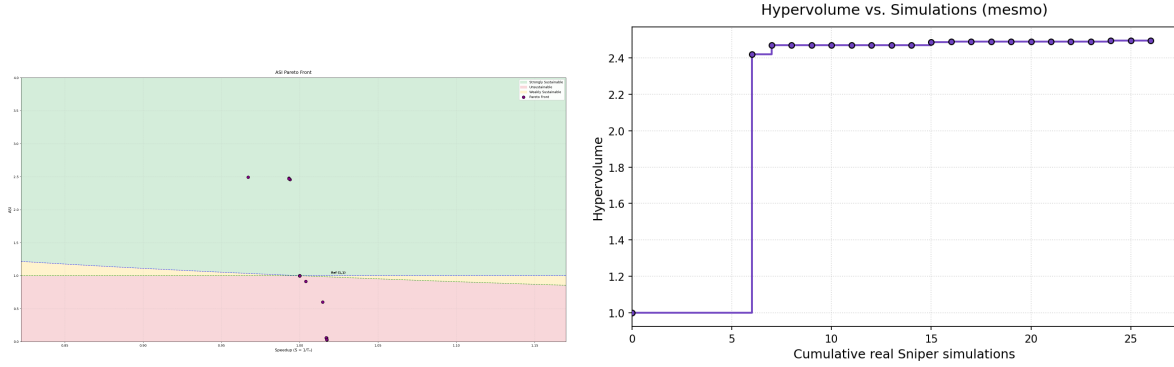
Final hypervolume: 2.7310

Front:

bpt	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m	4	64	4	32	8	512	8	2048			2	2	512	1.5862	1.0085	48.11	25.86	Strongly Sust.
pentium_m	4	64	4	32	8	512	4	1024			2	2	512	1.6928	1.0083	42.91	25.13	Strongly Sust.
pentium_m	4	64	4	32	8	512	4	1024			2	2	32	2.5911	1.0004	41.18	16.62	Strongly Sust.
pentium_m	4	16	4	32	8	512	4	1024			2	2	32	2.6146	0.9998	40.81	16.51	Strongly Sust.
nn	8	16	4	32	8	256	4	1024	64	0.005	2	2	64	2.6838	0.9823	37.99	16.39	Strongly Sust.
nn	8	16	4	32	8	256	4	1024	64	0.005	2	2	32	2.7184	0.9747	37.93	16.19	Strongly Sust.

3.1.2 hyperparameter settings 2

hyperparameter settings: 400 candidate samples are used and 5 initial configurations are ran. 20 iterations. 4 benchmarks: ML2, CCl, MIP and EI.



Result:

Total sniper runs: 26

Final hypervolume: 2.4945

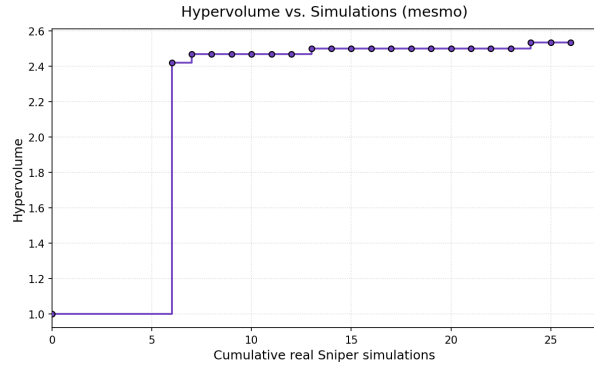
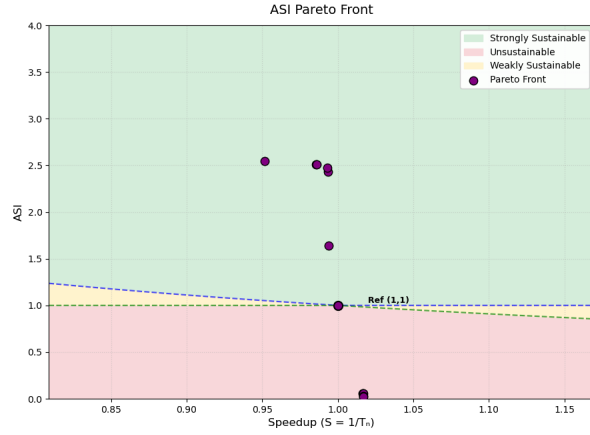
Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		4	64	4	16	8	512	8	4096				8	8	512	0.0267	1.0174	118.45	763.80	Unsustainable
a53	512	8	64	8	16	8	512	16	1024	2			10	8	512	0.0310	1.0172	107.21	763.19	Unsustainable
pentium_m		8	64	8	16	8	512	16	1024				10	8	256	0.0572	1.0171	83.61	535.26	Unsustainable
a53	512	8	64	8	16	8	512	16	1024	2			10	8	256	0.0572	1.0170	83.61	535.26	Unsustainable
a53	512	4	64	8	16	8	512	16	1024	2			10	8	256	0.0597	1.0170	79.19	534.47	Unsustainable
pentium_m		8	64	4	32	8	512	8	1024				2	10	64	0.6014	1.0147	63.70	60.61	Unsustainable
tage		8	64	4	16	4	512	8	1024				10	8	32	0.9122	1.0039	56.76	42.19	Unsustainable
baseline																1.0000	1.0000	94.01	27.56	Reference
pentium_m	4	64	8	16	4	256	4	2048					2	2	64	2.4560	0.9938	42.80	17.34	Strongly Sust.
pentium_m	4	16	8	32	4	256	4	2048					2	2	64	2.4675	0.9931	42.78	17.26	Strongly Sust.
pentium_m	4	16	8	16	4	256	4	2048					2	2	64	2.4779	0.9931	42.43	17.23	Strongly Sust.
nn	4	16	8	32	4	256	4	2048			64	0.005	2	2	64	2.4940	0.9671	42.78	17.07	Strongly Sust.

3.1.3 hyperparameter settings 3

hyperparameter settings: 400 candidate samples are used and 7 initial configurations are ran.

- 20 iterations



Result:

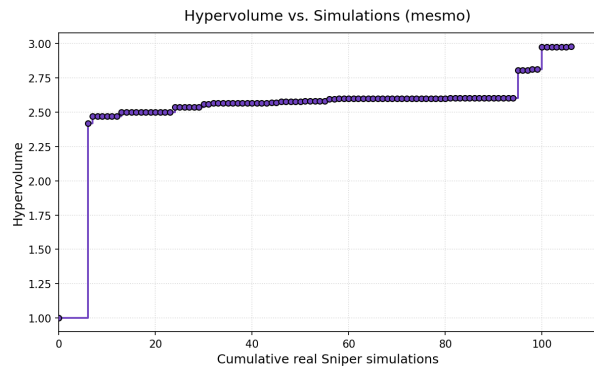
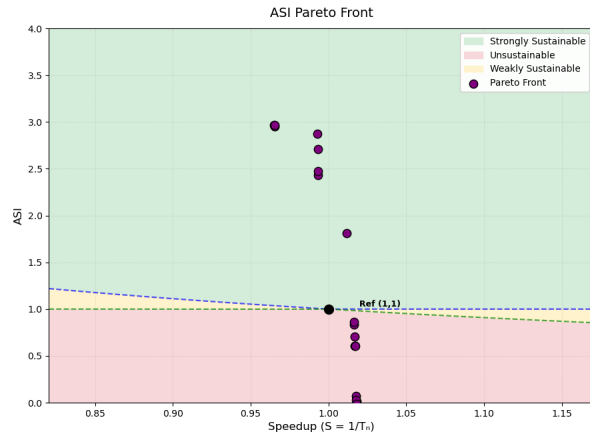
Total sniper runs: 26

Final hypervolume: 2.5351

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	2048	8	32	8	32	4	512	8	2048	4			8	10	256	0.0244	1.0170	105.00	997.11	Unsustainable
pentium_m		8	32	4	16	8	512	8	2048				8	8	512	0.0306	1.0169	108.38	762.95	Unsustainable
a53	1024	8	64	8	16	8	512	8	1024	3			10	8	256	0.0583	1.0169	81.53	535.21	Unsustainable
a53	1024	8	16	8	16	8	512	8	1024	3			10	8	256	0.0591	1.0163	80.08	535.08	Unsustainable
baseline																1.0000	1.0000	94.01	27.56	Reference
pentium_m		4	64	8	16	4	256	4	2048				2	2	512	1.6414	0.9940	44.46	25.64	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	256	2.4282	0.9933	42.61	17.56	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	64	2.4777	0.9933	42.43	17.23	Strongly Sust.
tage		4	16	8	16	4	256	4	2048				2	2	32	2.5100	0.9862	42.37	17.01	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	32	2.5104	0.9857	42.37	17.01	Strongly Sust.
nm		4	16	8	16	4	256	4	2048		64	0.001	2	2	32	2.5457	0.9516	42.37	16.77	Strongly Sust.

- 100 iterations



Result:

Total sniper runs: 106

Final hypervolume: 2.9794

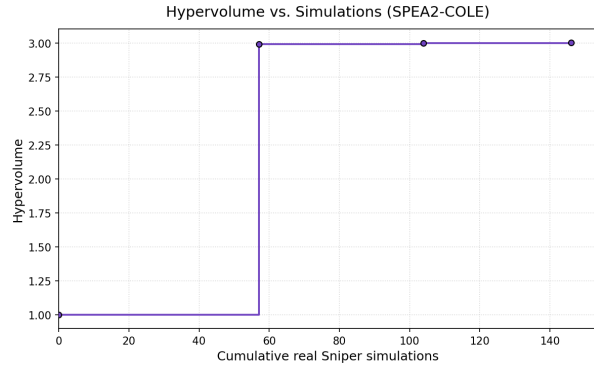
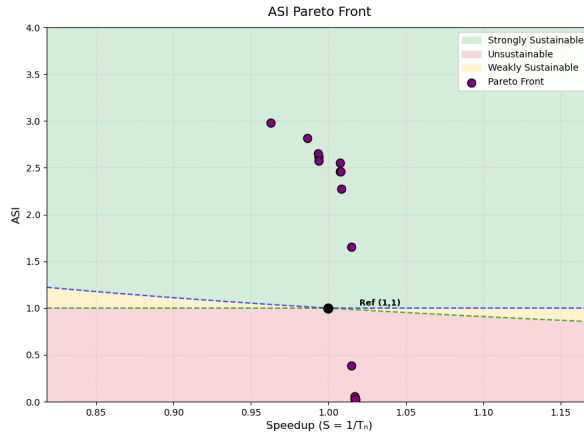
Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
tage		4	64	8	16	4	512	16	8192				8	10	1024	-0.0112	1.0180	288.40	2638.40	Unsustainable
pentium_m		4	64	4	64	8	512	8	1024				10	10	512	0.0115	1.0180	131.82	1434.20	Unsustainable
pentium_m		8	64	4	64	8	512	16	1024				8	10	256	0.0256	1.0177	101.17	995.75	Unsustainable
pentium_m		8	64	4	64	8	512	16	1024				8	10	128	0.0690	1.0175	78.77	464.46	Unsustainable
a53	2048	4	64	8	32	8	512	16	1024	4			8	10	64	0.6029	1.0171	63.08	60.76	Unsustainable
pentium_m		8	32	4	16	8	512	4	1024				8	10	64	0.6048	1.0169	63.13	60.55	Unsustainable
pentium_m		8	64	4	64	8	512	16	4096				8	8	64	0.7077	1.0167	77.22	45.90	Unsustainable
a53	512	4	32	8	64	8	512	4	2048	4			10	8	64	0.8359	1.0163	62.05	44.18	Unsustainable
a53	1024	4	32	4	32	4	512	16	2048	4			10	8	64	0.8658	1.0163	59.92	43.38	Unsustainable
a53	512	8	32	4	16	8	512	8	1024	2			2	4	64	1.8109	1.0115	48.09	22.65	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	256	2.4282	0.9933	42.61	17.56	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	2048				2	2	64	2.4777	0.9933	42.43	17.23	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	1024				2	2	64	2.7079	0.9931	35.48	16.52	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	2			2	2	64	2.8713	0.9929	34.06	15.72	Strongly Sust.
a53	512	4	64	4	16	4	256	4	1024	2			2	2	16	2.9494	0.9654	34.08	15.30	Strongly Sust.
a53	1024	4	32	4	16	4	256	4	1024	2			2	2	16	2.9640	0.9653	33.91	15.24	Strongly Sust.
a53	512	4	32	4	16	4	256	4	1024	2			2	2	16	2.9642	0.9651	33.91	15.24	Strongly Sust.

3.2 SPEA2 - COLE

3.2.1 Hyperparameter settings 1

Hyperparameter settings: Patience = 1 (amount of iterations to wait before stopping after no improvement), max_iterations: int = 10. (max iter never reached)



Result:

Total sniper runs: 146

Final hypervolume: 3.0042

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	2048	4	64	4	32	4	512	16	2048	3			8	10	512	0.0102	1.0176	138.09	1434.83	Unsustainable
a53	2048	4	64	4	32	4	512	16	2048	3			8	10	256	0.0256	1.0174	100.93	995.54	Unsustainable
pentium_m		4	32	4	32	8	512	16	2048				10	10	256	0.0257	1.0174	100.85	995.53	Unsustainable
a53	1024	8	64	8	32	8	512	8	1024	3			4	8	512	0.0317	1.0171	105.48	763.17	Unsustainable
tage		4	32	8	32	4	512	16	4096				10	10	128	0.0603	1.0170	92.12	466.61	Unsustainable
a53	1024	4	64	8	32	8	512	4	2048	3			8	4	512	0.3832	1.0150	60.14	97.85	Unsustainable
tage		8	64	4	32	4	512	8	2048				4	4	128	1.6527	1.0150	53.96	23.78	Strongly Sust.
pentium_m		4	64	4	64	4	512	16	2048				8	2	256	2.2741	1.0084	48.86	17.94	Strongly Sust.
pentium_m		4	64	4	32	8	512	16	1024				8	2	256	2.4611	1.0082	43.49	17.22	Strongly Sust.
pentium_m		4	16	4	64	8	512	16	1024				2	2	256	2.4626	1.0077	43.71	17.18	Strongly Sust.
a53	512	4	32	4	16	8	512	8	1024	3			4	2	128	2.5538	1.0074	40.89	16.89	Strongly Sust.
pentium_m		8	32	4	64	4	256	4	1024				4	2	256	2.5740	0.9936	39.32	16.94	Strongly Sust.
tage		4	64	8	16	8	256	4	1024				8	2	256	2.6172	0.9936	36.56	16.97	Strongly Sust.
pentium_m		4	16	8	16	4	256	4	1024				8	2	256	2.6515	0.9935	35.65	16.85	Strongly Sust.
a53	512	4	32	4	32	4	128	16	1024	3			4	2	256	2.8159	0.9865	35.58	15.87	Strongly Sust.
a53	512	4	16	4	32	8	128	16	1024	3			4	2	16	2.9808	0.9627	35.08	15.04	Strongly Sust.

4 ML2 benchmark, larger working set

Works with array where each element is 4 integers (32 bits) so 16 bytes. The size of the array used to be: $65536 * \text{sizeof}(el) = 65536 * 16 \text{ B} = 1048576 \text{ B} = 1 \text{ MiB}$

Pseudo random walk through this array.

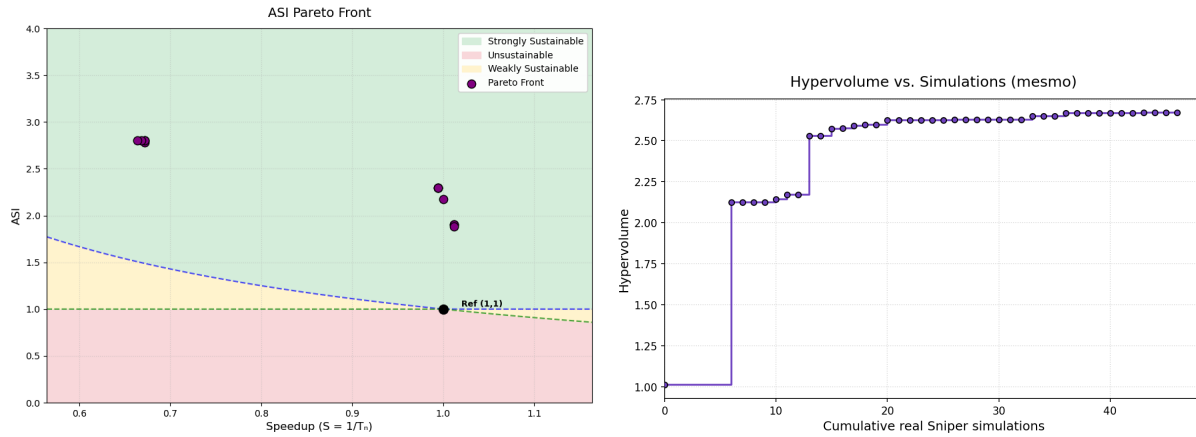
Now we adjust the amount of elements to $65536 * 4$ so we have 4 MiB

We expect each configuration where L3 is lower then working set to be really slow, and configuration with large enough L3 and maximum L2 size to be the fastest. (L3 size = 4 MiB already large enough so I assume L3 size = 8 MiB will not be on front because larger ASI)

The Baseline (reference) has L2 size 256 and L3 size 8192 so I assume speedup can never be extremely large, in next simulation we will change baseline to L3 size to not fit the working set.

All following simulations are done with MESMO

4.1 Reference: L3 size = 8 MiB, L2 size = 0.25 MiB



Result:

Total sniper runs: 46

Final hypervolume: 2.6719

I stopped after 40 iterations because we can see some points with L2 size to the maximum and L3 size 4 MiB have been found, aswell the complete opposite.

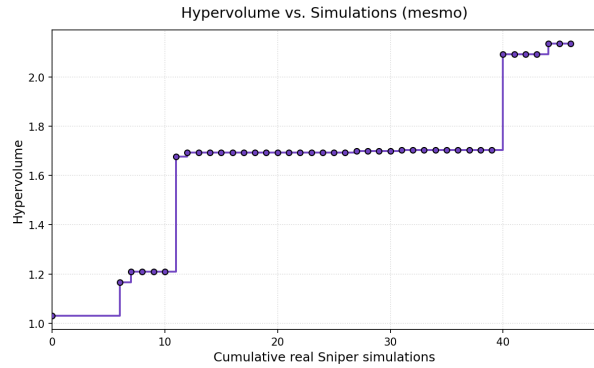
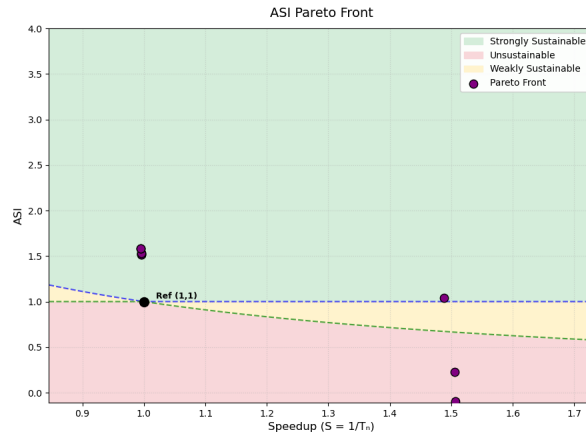
Right now I don't want to find complete front, I just want to see which clusters the search strategy can find.

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
pentium_m		8	64	4	16	4	512	4	4096				2	2	64	1.8803	1.0119	64.72	17.71	Strongly Sust.
tage		8	32	4	32	4	512	8	4096				8	2	32	1.9023	1.0117	64.54	17.53	Strongly Sust.
a53	2048	8	32	4	32	4	512	8	4096	4			8	2	32	1.9023	1.0117	64.54	17.53	Strongly Sust.
pentium_m		4	64	4	64	4	256	4	4096				4	2	128	2.1778	1.0002	54.51	16.55	Strongly Sust.
a53	512	4	64	4	64	8	128	8	4096	4			2	2	16	2.2985	0.9944	53.33	15.82	Strongly Sust.
nn		4	64	4	64	8	128	8	4096		16	0.001	2	2	16	2.2985	0.9941	53.33	15.82	Strongly Sust.
nn		4	64	4	64	8	128	8	4096		64	0.005	2	2	16	2.2986	0.9940	53.33	15.82	Strongly Sust.
pentium_m		4	64	8	64	4	256	8	2048				8	2	16	2.7839	0.6718	41.56	14.21	Strongly Sust.
pentium_m		4	64	8	16	4	256	8	2048				2	2	16	2.8017	0.6718	40.90	14.18	Strongly Sust.
tage		4	64	8	16	4	256	8	2048				2	2	16	2.8017	0.6718	40.90	14.18	Strongly Sust.
nn		4	64	8	16	4	128	4	2048		64	0.0005	10	2	32	2.8026	0.6683	41.76	14.09	Strongly Sust.
nn		4	64	8	16	4	128	4	2048		64	0.0005	8	2	32	2.8046	0.6641	41.76	14.08	Strongly Sust.

4.2 Reference: L3 size = 2 MiB, L2 size = 0.25 MiB

We can not compare HV to other searches with other reference points, but we can look what kind of Front we get and which configurations are found.



Result:

Total sniper runs: 46

Final hypervolume: 2.1357

Front:

bpt	bp	l1da	l1d	l1ia	l1i	l2a	l2	l3a	l3	nhist	nnbl	nnlr	robc	robd	robw	ASI	Speedup	Area	PeakPow	Region
a53	1024	8	32	8	16	8	512	8	8192	2			10	8	16	-0.0957	1.5062	108.88	43.97	Unsustainable
a53	1024	4	16	8	32	8	512	8	4096	2			10	8	64	0.2268	1.5059	73.96	43.61	Unsustainable
pentium_m		4	16	8	64	8	256	4	4096				10	2	16	1.0418	1.4885	55.92	16.49	Strongly Sust.
nn		8	64	4	32	8	128	16	2048		64	0.001	4	2	64	1.5155	0.9962	44.89	14.27	Strongly Sust.
a53	512	8	64	4	32	8	128	16	2048	4			4	2	64	1.5156	0.9961	44.89	14.27	Strongly Sust.
a53	512	8	32	4	32	8	128	16	2048	3			4	2	64	1.5297	0.9960	44.55	14.23	Strongly Sust.
nn		8	32	4	32	8	128	16	2048		32	0.005	4	2	64	1.5297	0.9958	44.55	14.23	Strongly Sust.
a53	512	8	32	4	32	8	128	4	2048	4			8	2	16	1.5796	0.9949	44.42	13.81	Strongly Sust.
nn		8	32	4	32	8	128	4	2048		32	0.005	8	2	16	1.5796	0.9948	44.42	13.81	Strongly Sust.

5 Benchmarks

5.1 ML2

ML2: L2 resident (depending on LFSR settings) linked list traversal.

Listing 1: ML2 benchmark source

```

1 #include <stdio.h>
2 #include <stdlib.h>      /* malloc, free, rand */

```



```

3
4 #include "common.h"
5
6 #define ASIZE 65536*4
7 #define STEP 128
8 #define ITERS 4096
9 #define LEN 2048
10
11
12 typedef struct dude {
13     int p1,p2,p3,p4;
14 } dude;
15
16
17 dude arr[ASIZE];
18 __attribute__((noinline))
19 int loop(int zero) {
20     int t = 0, count=0;
21
22     unsigned lfsr = 0xACE1u;
23     do
24     {
25         /* taps: 18 17 16 13; feedback polynomial:  $x^{18} + x^{17} + x^{16} + x^{13} + 1$  */
26         lfsr = (lfsr >> 1) ^ (-(lfsr & 1u) & 0x39000u);
27         lfsr = lfsr + arr[lfsr].p1;
28     } while(++count < ITERS);
29     while(lfsr != 0xACE1u);
30
31     return t;
32 }
33
34
35 int main(int argc, char* argv[]) {
36     argc&=10000;
37     ROI_BEGIN();
38     int t=loop(argc);
39     ROI_END();
40     volatile int a = t;
41 }

```

5.2 CCI

CCI: Impossible control with large basic blocks (potentially larger penalty)

Listing 2: CCI benchmark source

```

1     #include <stdio.h>
2 #include "randArr.h"
3 #include "common.h"
4
5 #define ASIZE 2048
6 #define STEP 256
7 #define ITERS 8
8
9 __attribute__((noinline))
10 int loop(int zero) {
11     int t = 0,i,iter;
12     for(iter=0; iter < ITERS; ++iter) {
13         for(i=0; i < STEP + zero; i+=1) {

```

```

14         if(randArr[i]) {
15             t+=3+3*t;
16             t&=0xFFF;
17             t+=3+3*t;
18             t&=0xFFF;
19             t+=3+3*t;
20             t&=0xFFF;
21             t+=3+3*t;
22             t&=0xFFF;
23             t+=3+3*t;
24             t&=0xFFF;
25             t+=3+3*t;
26             t&=0xFFF;
27             t+=3+3*t;
28             t&=0xFFF;
29             t+=3+3*t;
30             t&=0xFFF;
31             t+=3+3*t;
32             t&=0xFFF;
33         } else {
34             t&=0xFFF;
35             t+=3+3*t;
36             t&=0xFFF;
37             t+=3+3*t;
38             t&=0xFFF;
39             t+=3+3*t;
40             t&=0xFFF;
41             t+=3+3*t;
42             t&=0xFFF;
43             t+=3+3*t;
44             t&=0xFFF;
45             t+=3+3*t;
46             t&=0xFFF;
47             t+=3+3*t;
48             t&=0xFFF;
49             t+=3+3*t;
50             t&=0xFFF;
51             t+=3+3*t;
52         }
53     }
54 }
55 return t;
56 }
57
58 int main(int argc, char* argv[]) {
59     argc&=10000;
60     ROI_BEGIN();
61     int t=loop(argc);
62     ROI_END();
63     volatile int a = t;
64 }

```

5.3 MIP

MIP: Large Instruction Region – Instruction cache Misses

```

1     #include <stdio.h>
2     #include "common.h"

```

```

3
4 #define ASIZE 2048
5 #define STEP 128
6 #define ITERS 4
7
8 __attribute__((noinline))
9 int loop(int zero) {
10     int t = zero,i,iter;
11     for(iter=0; iter < ITERS; ++iter) {
12         t += 1;
13         t *= 2;
14         t -= 1;
15         t &= 0xFFF00FFF;
16         t ^= 0x14022;
17         t %= 0xFFFFFFF3;
18         t += 6;
19         t *= 10;
20         t += 1;
21         t *= 2;
22         t -= 1;
23         t &= 0xFFF00FFF;
24         t ^= 0x14022;
25         t %= 0xFFFFFFF3;
26         t += 6;
27         t *= 10;
28         t += 1;
29         t *= 2;
30         t -= 1;
31         t &= 0xFFF00FFF;
32         t ^= 0x14022;
33         t %= 0xFFFFFFF3;
34         ...
35         t += 1;
36         t *= 2;
37         t -= 1;
38         t &= 0xFFF00FFF;
39         t ^= 0x14022;
40         t %= 0xFFFFFFF3;
41         t += 6;
42         t *= 10;
43     }
44     return t;
45 }
46
47 int main(int argc, char* argv[]) {
48     argc&=10000;
49     ROI_BEGIN();
50     int t=loop(argc);
51     ROI_END();
52     volatile int a = t;
53 }

```

5.4 EI

EI: Integer Execution – 8 Independent computations per iteration

```

1     #include <stdio.h>
2     #include "common.h"

```

```

3
4 #define ASIZE 2048
5 #define STEP 128
6 #define ITERS 4096
7
8 __attribute__((noinline))
9 int loop(int zero) {
10     int t1 = 1;
11     int t2 = 89;
12     int t3 = 3;
13     int t4 = 21;
14     int t5 = 2;
15     int t6 = 7;
16     int t7 = 2;
17     int t8 = 3;
18     int t9 = 1;
19     int t10 = 89;
20     int t11 = 3;
21     int t12 = 21;
22     int t13 = 2;
23     int t14 = 7;
24     int t15 = 2;
25     int t16 = 3;
26
27     int i;
28
29     for(i=zero; i < ITERS; i+=1) {
30         t1+=i;
31         t2+=i;
32         t3+=i;
33         t4+=i;
34         t5+=i;
35         t6+=i;
36         t7+=i;
37         t8+=i;
38         t9+=i;
39         t10+=i;
40         t11+=i;
41         t12+=i;
42         t13+=i;
43         t14+=i;
44         t15+=i;
45         t16+=i;
46     }
47     return t1+t2+t3+t4+t5+t6+t7+t8*t9+t10+t11+t12+t13+t14+t15+t16;
48 }
49
50 int main(int argc, char* argv[]) {
51     argc&=10000;
52     ROI_BEGIN();
53     int t=loop(argc);
54     ROI_END();
55     volatile int a = t;
56 }

```